PROJECT BASED LEARNING

30 Data Analysis Questions

Student Practice Worksheet

Datasets: Match Results, Mobile Sales, and Fast Food

Instructions: Use Excel to clean, analyze, summarize, and visualize the data. Where appropriate, use formulas, PivotTables, charts, filters, sorting, and calculated columns. For the advanced questions, clearly explain the scoring method or assumptions you create.

Section 1: Match Results Dataset

1. Who are the top 10 batsmen based on their total runs scored across all matches?
2. Which 10 players have the highest strike rate among players who have faced at least 50 balls?
3. Which player has scored the most fours and which player has scored the most sixes?
4. Which team has won the highest number of matches, and how many matches did they win?
5. For each match, who was the highest run scorer? Can you display the player name, team, opponent, match date, and runs scored?
6. Do teams perform better while chasing or while batting first? Compare their average runs and overall performance in both situations.
7. Who are the top 10 bowlers based on total wickets taken, and what are their average economy rates?
8. Which players have scored the most runs through boundaries (fours and sixes), and how does their strike rate compare with other players?
9. Can you create a Player Performance Score using runs, strike rate, fours, sixes, and balls faced, and use it to rank the top 20 batsmen?
10. Can you combine the batting and bowling data to identify the top 10 all-round performers? Create your own Match Impact Score using runs, wickets, strike rate, economy rate, boundaries, and match result?

Section 2: Mobile Sales Dataset

1. Which mobile brand has the highest number of mobile models in the dataset?
2. Which are the top 10 mobile models based on their selling price?
3. What is the average selling price of mobiles for each brand?
4. Which mobile brand offers the highest average discount?
5. Which mobile brands have the highest average customer rating, and how does their average selling price compare?
6. How does mobile storage capacity affect the average selling price? Can you compare mobiles with different storage options?
7. Which mobile models provide the highest discount amount, and what percentage of their original price is discounted?
8. Which combination of memory and storage offers the best balance between selling price and customer rating?
9. Can you identify the top 10 mobile models that provide the best value for money by considering selling price, discount, storage, memory, and customer rating? Explain how you calculated the value score.
10. Can you analyze the factors that determine whether a mobile is classified as a "Good Product"? Which factors such as price, discount, rating, memory, or storage appear to have the strongest relationship with Good Product?

Section 3: Fast Food Dataset

1. Which restaurant has the highest number of food items in the dataset?
2. Which food items have the highest number of calories? Find the top 10 items.
3. What is the average calorie count of food items for each restaurant?
4. Which restaurant offers food items with the highest average protein content?
5. Which restaurant has the highest average amount of total fat, and which restaurant has the lowest?
6. Which food items contain the highest amount of sugar, and how many calories do those items contain?
7. How does the type of food item (salad or other) affect its average calories, fat, carbohydrates, and protein?
8. Which restaurant offers the best combination of low calories and high protein? Use the available nutritional information to support your answer.
9. Can you identify the 10 least healthy food items by creating a Health Risk Score using calories, total fat, saturated fat, trans fat, cholesterol, and sodium?
10. Can you compare the nutritional profiles of all restaurants and determine which restaurant provides the healthiest overall menu? Create your own scoring system using calories, fat, sugar, sodium, fiber, protein, and vitamins, and justify your ranking?